Chapter 2 — Discretisation

Error analysis and approximation theory

MTH3340 · Numerical Methods for PDEs

1 Error analysis of the Galerkin method

We have two ways of building $V_{N,0}$ (spectral, finite elements) and in both cases a family of solutions u_N. What is the error $u - u_N$, and how does it depend on N?

Definition (Standing assumptions)

$a : V_0 \times V_0 \to \mathbb{R}$ is bilinear, symmetric and positive definite, $\ell : V_0 \to \mathbb{R}$ is linear and continuous in the energy norm $\|\cdot\|_a$, and V_0 with the inner product a is a Hilbert space.

Both problems are then well posed (Chapter 1), and the discrete one inherits the assumptions because $V_{N,0} \subset V_0$ (conformity):

$$\begin{aligned} u \in V_0 \ &: \ a(u,v) = \ell(v) && \forall v \in V_0, \\ u_N \in V_{N,0} \ &: \ a(u_N, v_N) = \ell(v_N) && \forall v_N \in V_{N,0}. \end{aligned}$$

The analysis below is abstract: it never asks what $V_{N,0}$ is. The choice of space enters only at the very last step, through approximation theory.

Galerkin orthogonality

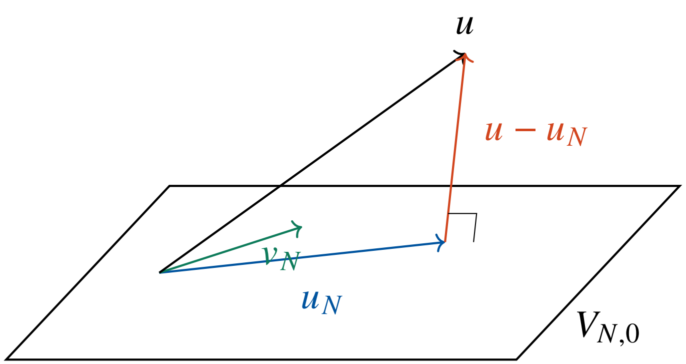
Since $V_{N,0} \subset V_0$, we may test the *continuous* problem with a *discrete* function v_N. Doing so in both problems and subtracting,

$$a(u - u_N, v_N) = 0 \qquad \forall v_N \in V_{N,0}.$$

Galerkin orthogonality: the error is orthogonal to the discrete space in the a-inner product,

$$u - u_N \;\perp_a\; V_{N,0}.$$

So u_N is the a-orthogonal projection of u onto $V_{N,0}$ — and orthogonal projections are best approximations. That is the next board.



Best approximation: Pythagoras

Pick any $v_N \in V_{N,0}$ and split $u - v_N = (u - u_N) + (u_N - v_N)$:

$$\|u - v_N\|_a^2 = \|u - u_N\|_a^2 + \|u_N - v_N\|_a^2 + \underbrace{2\, a(u - u_N,\ \underbrace{u_N - v_N}_{\in V_{N,0}})}_{=\,0\ \text{(orthogonality)}}.$$

The middle term is ≥ 0, so

$$\|u - u_N\|_a \;\leq\; \|u - v_N\|_a \qquad \forall v_N \in V_{N,0}.$$

Nothing in $V_{N,0}$ beats the Galerkin solution in the energy norm. The discretisation error is entirely an approximation question: how well can functions of $V_{N,0}$ approximate u at all?

Céa's lemma

Theorem (Céa's lemma)

Assume the Dirichlet data can be imposed exactly in the discrete space, i.e., the offset can be chosen as $u_{N,0} \in V_N$ with $u_{N,0} = g$ on Γ_0. Then the Galerkin approximation satisfies

$$\|u - u_N\|_a = \inf_{v_N \in V_{N,g}} \|u - v_N\|_a.$$

Proof. The hypothesis lets us use the same offset $u_{N,0}$ for both problems, so that

$$u \in u_{N,0} + V_0, \qquad u_N \in u_{N,0} + V_{N,0}.$$

Subtracting it, $\delta u \doteq u - u_{N,0} \in V_0$ and $\delta u_N \doteq u_N - u_{N,0} \in V_{N,0}$ solve

$$a(\delta u, v) = \tilde{\ell}(v)\ \forall v \in V_0, \qquad a(\delta u_N, v_N) = \tilde{\ell}(v_N)\ \forall v_N \in V_{N,0}, \qquad \tilde{\ell} \doteq \ell - a(u_{N,0}, \cdot):$$

a *homogeneous* problem and its Galerkin approximation on $V_{N,0} \subset V_0$, so the previous board applies verbatim. Finally $\delta u - \delta u_N = u - u_N$ and $u_{N,0} + V_{N,0} = V_{N,g}$. □

Consequence: if $V_{N,0} \subset V_{M,0}$ then $\|u - u_M\|_a \leq \|u - u_N\|_a$: raising p, or splitting every cell in two, can only improve the solution. But how fast?

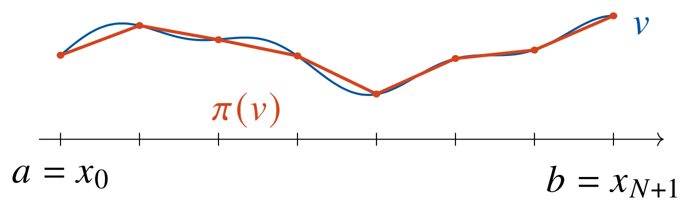
2 Approximation theory

Céa's lemma turned the error analysis into: find *one* good $v_N \in V_{N,0}$ and measure $\|u - v_N\|_a$. A convenient candidate is the interpolant of u.

Definition (Interpolation operator for 1D linear finite elements)

Given $V_N = \{v \in C^0([a,b]) : v|_{[x_{i-1},x_i]} \in \mathcal{P}_1\}$, define $\pi : V \to V_N$ by

$$\pi(v) \doteq \sum_{i=0}^{N+1} v(x_i)\, b_N^i.$$



This makes sense in 1D for $V = H^1((a,b))$ because H^1 functions are continuous in 1D, so the pointwise values $v(x_i)$ are meaningful.

Degrees of freedom and shape functions

The functionals $v \mapsto v(x_i)$ used by π are the degrees of freedom of the space.

Definition (Degrees of freedom)

A set of linear functionals $\{\sigma_i\}_{i=0}^{N+1}$ on V_N is an admissible basis of *degrees of freedom* if $I(v) \doteq [\sigma_0(v), \ldots, \sigma_{N+1}(v)]^T$ defines a bijection $I : V_N \longrightarrow \mathbb{R}^{N+2}$.

Definition (Shape functions)

Given a basis of degrees of freedom, the basis $\{b_N^0, \ldots, b_N^{N+1}\}$ of V_N is the basis of *shape functions* if

$$\sigma_i(b_N^j) = \delta_{ij}, \qquad i, j = 0, \ldots, N+1.$$

The basis of shape functions is unique, and for $\sigma_i(v) = v(x_i)$ it is precisely the hat function basis. With it, the interpolant is simply $\pi(v) = \sum_i \sigma_i(v)\, b_N^i$, and $\pi(v_N) = v_N$ for $v_N \in V_N$. DOFs + shape functions is the pattern we will generalise to 2D, 3D and higher order.

Stability of the interpolant

Lemma (Continuity of the 1D interpolant)

The map $\pi : H^1(\Omega) \to V_N \subset H^1(\Omega)$ is bounded uniformly in the mesh width h, for bounded $\Omega \subset \mathbb{R}$: $\|\pi(v)\|_{H^1(\Omega)} \le c\|v\|_{H^1(\Omega)}$.

Sketch. For $v \in H^1$ and $x, y \in \overline{\Omega}$, Cauchy–Schwarz gives

$$|v(y) - v(x)| \le \int_x^y |v'| \le |y-x|^{\frac{1}{2}} |v|_{H^1(\Omega)}.$$

Seminorm. On a cell, $\pi(v)'|_{[x_{i-1},x_i]} = \frac{v(x_i)-v(x_{i-1})}{h_i}$, so the display applied on $[x_{i-1}, x_i]$ gives $|\pi(v)|_{H^1([x_{i-1},x_i])} \le |v|_{H^1([x_{i-1},x_i])}$; adding over cells, $|\pi(v)|_{H^1(\Omega)} \le |v|_{H^1(\Omega)}$.
L^2 norm. Now take $x = x_0$, a *minimum point of* $|v|$ on $\overline{\Omega}$: then $|v(x_0)|^2|\Omega| \le \int_\Omega |v|^2$, and the triangle inequality controls pointwise values,

$$\|v\|_{\infty,\Omega} \le |\Omega|^{-\frac{1}{2}} \|v\|_{L^2} + |\Omega|^{\frac{1}{2}} |v|_{H^1}.$$

Since $\|\pi(v)\|_{L^2} \le |\Omega|^{\frac{1}{2}} \|\pi(v)\|_\infty \le |\Omega|^{\frac{1}{2}} \|v\|_\infty$, this gives $\|\pi(v)\|_{L^2} \le \|v\|_{L^2} + |\Omega|\,|v|_{H^1}$. □

The constant does not depend on h, which is what makes the estimate useful along the whole family of meshes.

Interpolation error

Definition (The space $H^2(\Omega)$)

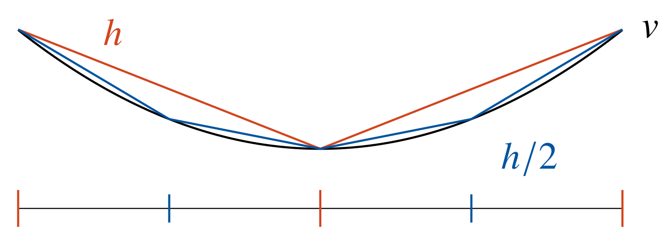
$H^2(\Omega) \doteq \{v \in H^1(\Omega) : v' \in H^1(\Omega)\}$, with seminorm $|v|_{H^2(\Omega)} \doteq |v'|_{H^1(\Omega)}$.

Lemma (Error bounds for the 1D interpolant)

For all $v \in H^2(\Omega)$ and a mesh of width h,

$$\|v - \pi(v)\|_{L^2(\Omega)} \le h^2 |v|_{H^2(\Omega)}, \qquad |v - \pi(v)|_{H^1(\Omega)} \le h\,|v|_{H^2(\Omega)}.$$

Sketch. On a cell, $v - \pi(v)$ vanishes at both end-points, so Poincaré–Friedrichs gives $\|v - \pi(v)\|_{L^2} \le h_i |v - \pi(v)|_{H^1}$. The derivative $v' - \pi(v)'$ vanishes somewhere in the cell (mean value theorem) and $\pi(v)'' = 0$ there, so the same bound applies to it. Combine and add over cells. □



Where H^2 bites: $v \in H^2 \Rightarrow v' \in H^1$, continuous in 1D — the same embedding, one derivative up, and what the mean value theorem needs.

The a priori error estimate

Chain Céa's lemma with the interpolation bound, taking $v_N = \pi(u)$:

$$\|u-u_N\|_{H^1(\Omega)} \simeq \|u-u_N\|_a \leq \|u-\pi(u)\|_a \simeq \|u-\pi(u)\|_{H^1(\Omega)} \leq Ch\,|u|_{H^2(\Omega)}.$$

Corollary (Error estimates for the 1D finite element solution)

Under the conditions of Céa's lemma, for a linear finite element space on a mesh of width h and $u \in H^2(\Omega)$,

$$\|u-u_N\|_{H^1(\Omega)} \leq Ch\,|u|_{H^2(\Omega)}, \qquad \|u-u_N\|_{L^2(\Omega)} \leq Ch^2|u|_{H^2(\Omega)}.$$

The L^2 estimate needs a *duality* argument (out of the scope of the course): weaker norm, one extra order.

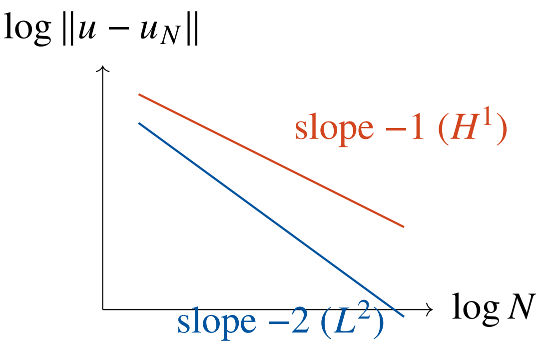
The energy norm is equivalent to the H^1_0 norm by coercivity and continuity of a, which is what lets us move between $\|\cdot\|_a$ and $\|\cdot\|_{H^1}$ above.

Error versus cost

On a uniform mesh, $h = \frac{|b-a|}{N+1}$, so

$$N \simeq \frac{1}{h}, \qquad \|u - u_N\|_{H^1} \le Ch \simeq \frac{C}{N}.$$

Halving h halves the error in H^1, divides it by four in L^2, and doubles the size of the linear system. That is the deal we announced: accuracy bought with cost.



The regularity assumption is not free: if $u \notin H^2$ (re-entrant corners, jumping coefficients) the rate degrades.

Q: Can we get a better rate without shrinking h?

Yes: raise the polynomial order.

Higher order finite elements

Use polynomials of degree p on each cell,

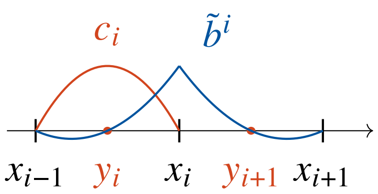
$$V_N^p \doteq \left\{ v \in C^0([a,b]) \ : \ v|_{[x_{i-1},x_i]} \in \mathcal{P}_p([x_{i-1},x_i]) \right\}.$$

The mesh nodes no longer determine such functions, so we add degrees of freedom,

$$\begin{aligned} \sigma_i^1(v) &\doteq v(x_i), & i &= 0,\dots,N+1, \\ \sigma_i^k(v) &\doteq v\Big(x_{i-1} + \tfrac{k}{p}(x_i - x_{i-1})\Big), & i &= 1,\dots,N+1,\ k = 1,\dots,p-1, \end{aligned}$$

i.e., $p-1$ equispaced interior nodes per cell, and $\dim V_N^p = p(N+1)+1$.

Locality survives: interior (bubble) shape functions live on a single cell, vertex shape functions on two.



h or p?

Lemma (Error bounds for the 1D interpolant, arbitrary order)

For $v \in H^{p+1}(\Omega)$ and the order-p interpolant π^p on a mesh of width h,

$$\|v-\pi^p(v)\|_{L^2(\Omega)} \le Ch^{p+1}|v|_{H^{p+1}(\Omega)}, \qquad |v-\pi^p(v)|_{H^1(\Omega)} \le Ch^p|v|_{H^{p+1}(\Omega)},$$

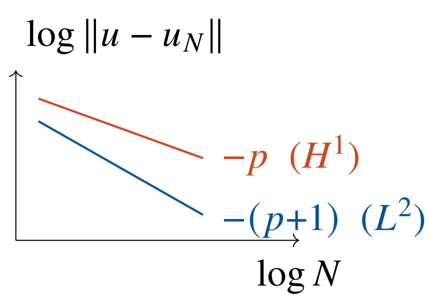
with C independent of h and v.

Corollary (Error estimates for the 1D finite element solution, order p)

Under Céa's conditions, for V_N^p and $u \in H^{q+1}(\Omega)$, $0 < q \le p$,

$$\|u-u_N\|_{H^1(\Omega)} \le Ch^q|u|_{H^{q+1}(\Omega)}.$$

Two knobs, one estimate: $h \downarrow$ ($N \uparrow$) or $p \uparrow$. The order is capped by the regularity of u ($q \le p$).



Raise p for smooth solutions, refine where the solution is rough.